accomplished quite simply in the following manner: Radioactive atoms of one of the elements contained in the poisonous substances are added to the latter, and the substances are then sent through the filter. These radioactive atoms undergo the same chemical reactions as the stable ones. After the poisonous substances have passed through the filter, one merely has to determine whether or not radioactivity appears at the other end of the filter, and if so, the intensity of this radioactivity will indicate what percentage of the poisonous substance has passed through the filter. The individual parts of the filter can also be checked, since, after the poisonous substance has passed through, the intensity of the radioactivity acquired by the parts in question by adsorption of the poisonous substance can be determined. This procedure will furnish information concerning the efficiency of the individual parts of the filter. Similarly, it is possible to check whether or not the rubber covering of the gas mask is actually hermetically sealed against the poison gas, by bringing the latter in contact with one side of the rubber covering and observing whether any radioactivity can be detected on the other side. If so, the rubber covering is proved to be unsuitable for a gas mask. Such testing procedures have been described by Born and Zimmer, and are actually being used.

VI. ARTIFICIAL RADIOACTIVE SUBSTANCES IN CHEMISTRY

In chemistry, artificial radioactive substances are used as tracers on an ever increasing scale. Let us give first an example from the field of quantitative analysis: Erbacher and Philipp attempted a quantitative analysis of a mixture of gold, iridium and platinum. After a reduction by hydrogen peroxide, the gold was precipitated in its metallic form, and was then weighed in order to determine whether its quantity was sufficiently close to the original quantity of gold introduced into the mixture—in other words, whether the gold has actually been separated quantitatively. This actually seemed to be the case—the separation appeared to have been completely successful. As a countercheck, a small amount of radioactive gold was added to the gold. It was found that the radioactivity of the separated quantity was perceptibly less than that of the original quantity. This proved that the gold had not been separated quantitatively, and that the seemingly perfect result had been due to the fact that a quantity of platinum and iridium had accompanied the precipitated gold, and the quantity of this platinum and iridium happened to be exactly equal to the missing quantity of gold.

This example reveals what the important factor is in the utilization of radioactive substances in quantitative analysis. A small quantity of a radioactive isotope of a substance is added, as a tracer—as a label or tag, as it were—to the substance itself. Then, the radioactivity of the tracer enables us